

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Requisition for Barred Tees
Document Number	EA-BFPCL-GPMC-GGPL-FEED-PPL-REQ-002
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Requisition for Barred Tees

R01	08.08.22	Issued for Review	I. Osobase	E.Nwachukwu	A.Duru	
Rev #	Issue Date	Status Description	Originator	Reviewer	Approver (Epic)	Approver (BFPCL)

Revision History

Rev No.	Date	Description of Revision
R01	08.08.22	Issued for Review

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED-PPL-REQ-002

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviations	Meaning
CIF	Cost, Insurance and Freight
CFR	Cost and Freight
EPC	Engineering, Procurement and Construction
NDT / NDE	Non-Destructive Testing / Non-Destructive Examination
ITP	Inspection and Testing Plan
PO	Purchase Order
QA/QC	Quality Assurance and Quality Control
VPIS	Vendor Print Index & Schedule
UOM	Unit of Measurement
CIF	Cost, Insurance and Freight
RFQ	Request for Quotation
N/A	Not Applicable

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

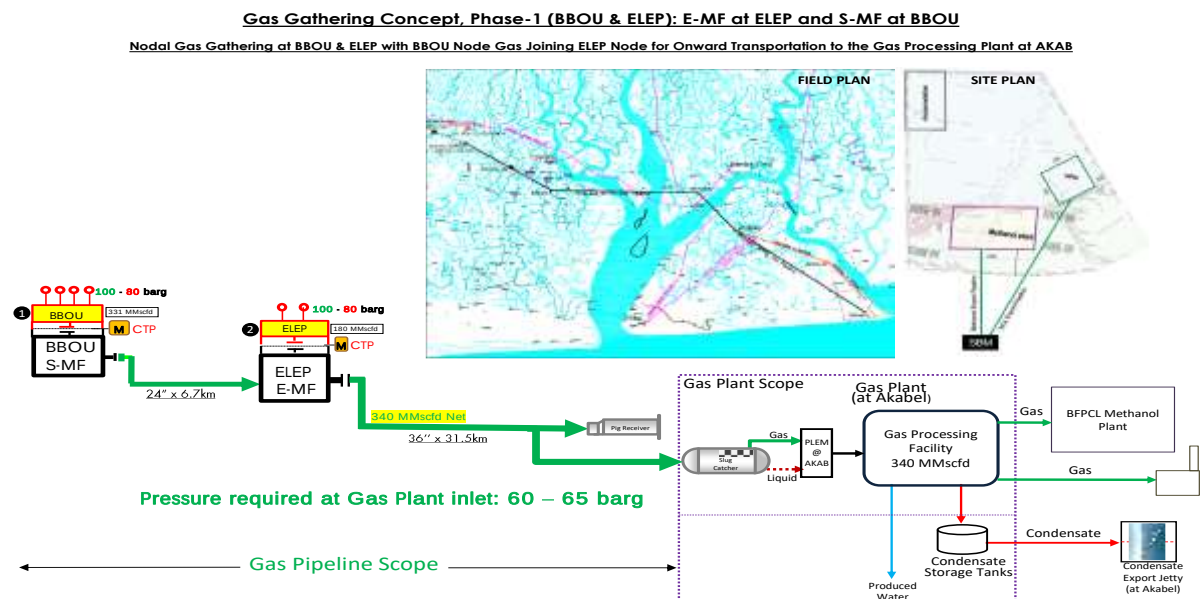


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

This Material Requisition covers the scope of supply of Barred Tee to be used above ground on the Pipeline systems of the BFPCL Gas Gathering Project-Phase 1. The barred Tees shall be installed at BBOU Manifold Platform, ELEPA Manifold Platform and AKABEL Pigging Stations.

This document shall be read in conjunction with the Project Specifications, Reference Documents, Datasheets and related Industry Codes and Standards.

1.3. Reference Project Documents

S/N	Document number	Title
1	EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis for Design Document
2	EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-008	Specification for Barred Tee
3	EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-008	Data Sheet for Barred Tee
4	DEP 31.40.00.10-Gen	Pipeline Engineering (Amendments/Supplements to ISO 13623)
5	DEP 31.40.10.13-Gen	Design of Pipeline Pig Trap Systems
6	ASTM A694	Standard Specification for Carbon and Alloy Steel Forgings for Pipe Flanges, Fittings, Valves and Parts for High-Pressure Transmission Service
7	MSS SP-75	Specification for High-Test, Wrought, Butt-Welding Fittings
8	ASME B31.8	Gas Transmission and Distribution Piping Systems
9	MSS SP 25	Standard Marking System for Valves, Fittings, and Unions

2. Definition of Terms

Terms	Definitions
Project	Gas Gathering Pipelines & Associated Infrastructure (Phase1)
Company	Brass Fertilizer & Petrochemical Limited (BFPCL) that initiates the project and ultimately pays for its design Procurement and construction. The Employer will generally specify the technical requirements. The Employer may also include an agent or consultant authorized to act for and on behalf of, the Employer.
EPC Contractor	The party who Company has employed to perform all works related to the Scope of Work of the Project.
Engineer	The party that carries out the FEED Engineering of the project.
Manufacturer / Vendor / Supplier	Company or factory responsible for design and manufacture of Equipment and provides it to the Owner of this Project.
Sub-Vendor	Company or factory that provides part of or complete services in the design, manufacture and supply of the equipment on behalf of the Vendor for this Project.
Shall	The word 'shall' indicate a mandatory requirement.
Should	The word 'should' indicate a recommendation.

3. Scope of Supply

- Vendor's scope of work includes but not limited to Sourcing of Material, Manufacture, Supply, Inspection and Testing, Painting and Coating, all necessary quality control and quality assurance incidental to the manufacture and supply of the Barred Tees Listed in Attachment 1- MTO for Barred Teed, producing required documentation as per the approved VPIS, handling, storage and shipping and providing total performance guarantee of the Barred Tee, in accordance with Project Specifications, Datasheets, international codes, standards and all applicable documents.
- VENDOR shall provide all the equipment, labour, materials, and consumables required for the design, fabrication, and supply of materials, inspection, testing, painting, supply and delivery of the Barred Tee.

Table 1 - Scope of Supply

Item No.	Tag No.	Item Description	UOM	Quantity
1.	EQUIPMENT / MATERIAL			
1.1	Stamping shall be as per applicable specifications and data sheets	Barred Tee The scope of supply shall include: Engineering, procurement, fabrication, inspection and testing, surface preparation and painting, documentation, preparation for shipment, preservation for supply of Barred Tee in compliance with this MR, Project Specifications, Data Sheet and other referenced documents. The Barred Tee within the scope of this supply are listed in the reference document (MTO for Barred Tee.) in Attachment 1 of this MR. For detailed scope, refer to - EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-008 Specification for Barred Tee - EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-008 Data Sheet for Barred Tee	Nos.	Lots
2	INSTALLATION AND COMMISSIONING SPARE PARTS			
	- Spare parts shall be provided for successful commissioning and start-up of supplied goods, and for Two years operation and maintenance. - Special tools if any shall be supplied for Successful commissioning and start-up of supplied goods, and for operation and maintenance. - Documentations for special tools and spare parts to be updated with final documentation-	N/A N/A N/A		
3	TESTING AND INSPECTION			

3.1	-	Inspection and Testing shall comply with the requirements of the under listed document and Industry Standard 3. EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-008 Specification for Barred Tee 4. EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-008 Data Sheet for Barred Tee	Lot	1
3.2	-	All Supplier/ Material Certificates Required as per Project Specification and Industry Standard	Lot	1
4	OTHER SERVICES (NON-MATERIAL DELIVERABLES)			
4.1	-	Engineering Services, as per requisition and Project specification.	Lot	1
4.2	-	Submission of Documents as per VPIS in Attachment-B	Lot	1
4.3	-	Surface preparation, painting etc., as applicable based on Company Approved Specification for Painting & Coating System.	Lot	1
4.4	-	Preservation Suitable for seaworthy and road transport and multiple handling. EA-BFPCL-GPMC-GGPL-FEED-MPP-PRC-001; Equipment & Material Preservation Procedure. EA-BFPCL-GPMC-GGPL-FEED-QAC-PLN-001; Project Quality Management Plan (QA & QC)	Lot	1
4.5	-	Shipping Documents and Customs Clearance as per project requirements.	Lot	1
4.6	-	Meetings requirements (Kick-off, Progress, Vendor document review and TQs etc.)	Lot	1
4.7	-	Warranties/ guarantee as per applicable Specifications of Barred Tee	Lot	1
4.8	-	Estimated delivery term is CIF based on INCOTERMS 2020. (The delivery terms will be finalized before placement of PO).	Lot	1
4.9	-	All Barred Tee supplied shall be from Owner Approved Vendor & Sub Vendor List.	Lot	1
4.10	-	PMI for SS material	N/A	1
5	OPTIONAL ITEMS			

5.1		Delivery CFR Nigerian Port basis	Lot	1
6	LIST OF EXCLUSIONS			
6.1		Supply of Anchor bolts	N / A	
6.2		Installation at site / Erection	N / A	1
6.3		Commissioning	N / A	1
6.4		Foundation Design	N / A	1

3.1. Technical Notes

All technical notes shall be read in conjunction with the following documents

- EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-008; Specification for Barred Tee
- EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-008; Data Sheet for Barred Tee

4. Services

4.1. General

- Design, procurement-inspection, fabrication-inspection, Documentation, Drawings, Certification, of the Barred Tee
- The final design of Barred Tee shall be the sole responsibility of the VENDOR. The VENDOR shall ensure that the final design of the Barred Tee complies fully with the design criteria as per project requirement and applicable specifications and data sheets.
- VENDOR shall also be responsible for the procurement of all materials required for the fabrication of the Barred Tee, design certification, packing and transportation of Barred Tee and all necessary accessories
- VENDOR shall develop General Arrangement drawing for all the Barred Tee within the scope of supply and issue to client for review and approval prior to fabrication

4.2. Project management

VENDOR shall include the dedicated Project Team which coordinates all the engineering services throughout the project cycle including establishment of overall Project Execution plan compliance with contractual, statutory & regulatory requirements and Technical/commercial logistic requirements.

4.3. Engineering Services

The VENDOR is responsible for the procurement, fabrication, inspection/testing, supply, guarantee and documentation of all the Barred Tee. This shall include, but not limited to:

- Design, Engineering for the complete supply of materials
- Drawings and Documentation.
- Resolving engineering technical queries relating to equipment within the scope of supply.
- Co-ordination with EPC Contractor for detailed design of all the components within the scope of supply.
- Ensuring that all material supplied shall meet the requirements of the datasheets, specifications, design codes and engineering standards referred herein.
- Expediting services for the sub-contracted items. Ensuring that sub VENDORS shall adhere to project requirements specifications and datasheet.
- Participating in co-ordination meetings and kick-off meeting.

4.4. Supply of spares as per Project Scope.

N/A

5. Information Required along with Bid

VENDOR shall submit all the component details including applicable Specification for the Barred Tee, Datasheet and Drawings completed with vendor offer, and other project documents in their technical offer.

VENDOR shall include in the technical bid, his previous experience in the provision of similar service for the same design and rated conditions.

5.1. Deviations and Alternatives

- The deviation list shall be prepared by VENDOR and shall be included in Vendor's bid/proposal (stamped and signed scan copy). in any case where VENDOR has no exceptions, VENDOR shall stamp and sign clearly stating "No Deviations" in VENDOR deviation form, Technical explanations for proposed deviations shall be included in VENDOR deviation form as per Attachment D.
- VENDOR shall be responsible for any deviation or alternatives included in VENDOR'S bid/proposal and not stated in the deviation list. Any such hidden deviations shall not be entertained. Vendor is requested to submit filled in "Deviation List" and a completed technical checklist in the format attached in Attachment D and E.
- Assumptions, Alternatives, Deviations, Exceptions and Concessions raised shall be recorded and supplemented with adequate technical justifications. If no deviation list in the required format is furnished with the Bid, then it shall be construed that Vendor has no deviations to the requisition and its attachments and the Bid is in full compliance.
- After placement of Purchase Order, no deviation shall be acceptable from VENDOR.
- Vendor to provide the filled-out form as per Attachment C and Completed technical checklist along with bid.

ATTACHMENT A - Technical Documents and Attachments

Table 2 - Technical Documents

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PPL-SPC-008	Specification for Barred Tee
EA-BFPCL-GPMC-GGPL-FEED-PPL-DSH-008	Data Sheet for Barred Tee
EA-BFPCL-GPMC-GGPL-FEED-MPP-PRC-001	Equipment & Material Preservation Procedure.
EA-BFPCL-GPMC-GGPL-FEED-QAC-PLN-001	Project Quality Management Plan (QA & QC)
ATTACHMENT 1	MTOs for Barred Tee

ATTACHMENT B - Vendor Drawing and Data Requirement List

Table 3 - Vendor Drawings and Data Requirement List

Sr. No.	Drawing & Data Required	Electronic Copies required		Due Date **	Required for Final submission	Due Date	Nos of Copies	Due Date
		For Bid	For Approval *				For Data Book	
1.	Vendor Print Index & Schedule	Y	Y	2 Weeks	Y		3C + 3E Flash-Driver / HardDisk	
2.	Vendor / Manufacturer's Catalogue	Y	Y	2 Weeks	Y		3C + 3E Flash Driver / HardDisk	
4.	Fabrication/Delivery Schedule		Y	4 Weeks	Y		3C + 3E Flash-Driver / HardDisk	
5.	Project Quality Plan, QAC Procedures including	Y	Y	4 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
6.	Manufacturing Procedure Specification (MPS)	Y	Y	2 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
7.	Manufacturing Procedure Specification (MPS)		Y	2 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
8.	Material Handling, Storage, Stacking And Shipping /Transportation Procedures		Y	3 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
9.	Certified Hydro-static Test Report		Y	1 week before inspection release note	Y		3C + 3E Flash Driver / Hard Disk	

Sr. No.	Drawing & Data Required	Electronic Copies required		Due Date **	Required for Final submission	Due Date	Nos of Copies	Due Date
		For Bid	For Approval *				For Data Book	
10.	Material Test Reports		Y	1 week before inspection release note	Y		3C + 3E Flash Driver / Hard Disk	
11.	NDT/ NDE Test Reports		Y	1 week before inspection release note	Y		3C + 3E Flash Driver / Hard Disk	
12	Inspection and Test Plan	Y	Y	2 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
13.	General Arrangement Drawing	Y	Y	2 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
14	Deviations	Y		With Bid			3C + 3E Flash Driver / Hard Disk	
15	Progress Reports		Y	Weekly			3C + 3E Flash Driver / Hard Disk	
16	Sub Vendor List	Y	Y	2 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
17	Preservation at Site		Y	2 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
18.	PWHT Procedure, as applicable		Y	2 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
19	Certification Data Book		Y	2 Weeks	Y		3C + 3E Flash Driver / Hard Disk	

Sr. No.	Drawing & Data Required	Electronic Copies required		Due Date **	Required for Final submission	Due Date	Nos of Copies	Due Date
		For Bid	For Approval *				For Data Book	
20	External Painting Procedure/ Specification		Y	4 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
21	Material test certificate as per EN 10204 as mentioned in Data Sheet		Y	4 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
22	Certified Welding Procedure Specification (WPS), Procedure Qualifications (WPQ) and Welders Performance Qualifications Record(WPQ)		Y	2 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
23	Detail Drawings of Equipment's with Bill of Materials		Y	2 Weeks	Y		3C + 3E Flash Driver / Hard Disk	
24	Shipping Schedule and Packing List		Y	2 Weeks	Y		3C + 3E Flash Driver / Hard Disk	

Sr. No.	Drawing & Data Re- quired	Electronic Copies required		Due Date **	Required for Final submis- sion	Due Date	Nos of Copies	Due Date
		For Bid	For Approval *				For Data Book	
* After P.O. ** From P.O. Date Legends: C: Copy E: Electronic Copy Note: 1. The above-mentioned documents are general requirements; the same shall be discussed and agreed with Company / its representative and Vendor, prior to Placement of Order. 2. Electronic copy required at all stages. 3. One copy of IOM document shall be submitted along with Package delivery.								

ATTACHMENT C - Table of Compliance

VENDOR shall confirm below the Sections of the Requisition/Specification that he has complied with/accepted or further clarifications are desired, or VENDOR has taken a deviation with reasons thereto. Columns 2, 3, and 4 must be answered and initialed.

Table 4 - Table of Compliance

1	2	3	4
Specification/Datasheet No.	*Conforms Yes/ No	*Deviations Yes/ No	@Clarification or Reasons for Deviations
Specification for Flanges: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-013			
Datasheet for Pipes and Fittings: EA-BFPCL-GPMC-GGPL-FEED-MPP-SPC-010			
Equipment & Material Preservation Procedure: EA-BFPCL-GPMC-GGPL-FEED-MPP-PRC-001			
Project Quality Management Plan (QA & QC): EA-BFPCL-GPMC-GGPL-FEED-QAC-PLN-001			

Notes:

- Acceptance (i.e. yes) implies total compliance to the Specifications.
- Provide additional explanatory sheets if required.
- Above are the main documents to be complied. However, compliance is required on all documents referenced in this requisition.

ATTACHMENT D - Deviation List

Table 5 - Deviation List

Sr. No	Docu-ment	Refer-ence / Par-agraph	Devia-tion / Clari-fication	Justification for Devia-tion / Clarifi-cation	EPC Con-tractor Ap-proval	OWNER Approval	Remarks

Notes:

In case of no deviations, VENDOR shall state "NO DEVIATIONS" with stamp and sign below.

- Deviations listed elsewhere in VENDOR's offer shall not be considered. VENDOR shall be responsible to comply at his own cost and time.
- Offers that do not include signed and stamped Deviation List shall be liable for rejection.
- No deviations from any part of this requisition are allowed unless stated in VENDOR's bid using the above format.

ATTACHMENT E -Technical Checklist

VENDOR shall ensure that the following minimum information is attached with their bid.

Table 6 - Technical Checklist

S. NO	DESCRIPTION	PLEASE REPLY YES/ NO
•	VENDOR to furnish Quotation Reference and Quotation	
•	Completely filled in 'Table of Compliance'.	
•	Deviations, if any, to the scope of work, specification, codes and standards attached / referred to in this requisition with proper reasoning.	
•	Completely filled in data sheets.	
•	Signed and Stamped Specifications	
•	VENDOR's quality control documents (viz. Manufacturing, inspection and testing)	
•	Compliance that delivery schedules and VENDOR's detailed order execution schedule is submitted.	
•	QA/QC documents as required with bid.	
•	List of Special Tools and Tackles (if applicable)	
•	Bidder to Guarantee the performance parameters as per the specifications and datasheets.	
•	Bidder to confirm that quality certificate of the manufacturer facility and quality plans/ inspection and test plans is furnished with the bid.	
•	Any additional documents/data as required by bidder.	
•	SUPPLIER to quote for all items (type & design etc.) as per material requisition scope of supply and duly filled in signed and stamped and attached with the Offer.	
•	SUPPLIER to confirm that the offer is in full compliance with all requirements of Inquiry notes, Specifications / Standards attached with Inquiry, MTO for Barred Tee as per Attachment-1	
•	SUPPLIER to confirm that Unit price, total price and delivery period has been filled in, in the Material Requisition Scope of Supply.	

•	SUPPLIER to confirm that quotation shall be applicable and valid for placement of order either in full or in part based on the unit rates quoted.	
•	SUPPLIER to confirm that shipping weights and volume has been mentioned in the offer.	
•	SUPPLIER to confirm that they are aware of all arrangements for testing and inspection, in compliance with the Inquiry/specification, at SUPPLIER's work shall be done by them.	
•	SUPPLIER to indicate Spare part(s) for Two years continuous Operation, Commissioning spare(s) & Maintenance spare(s) (Itemized price list) if any	Not Applicable
•	SUPPLIER to confirm that they have indicated the country of origin and is aware that the certificate of origin shall be part of shipping documents.	
•	SUPPLIER shall indicate the Production / Delivery schedule.	
•	SUPPLIER to confirm that they have attached technical catalogue or product brochure with the offer.	
•	SUPPLIER to attach their "Quality Management System" along with the quotation.	
•	SUPPLIER to confirm that the typical ITP (Inspection and Test Plan) for the quoted items has been attached along with the quotation.	
•	SUPPLIER to confirm that Certificates for ISO accreditation for Self and Sub SUPPLIERS are attached with offer	N/A
•	SUPPLIER to furnish the reference list for the past jobs executed with other customers along with customer name, project name, Items supplied qty of orders etc.	
•	SUPPLIER to state the warranty period and confirm that the materials shall be replaced at free of cost in case any manufacture/ transit defects found during that period.	
•	SUPPLIER to confirm that the components quoted by them are suitable for the life span of 30 years.	
•	Manufacturing/ Production Schedule.	
•	Tag numbers shall be as per applicable Specification or as provided by Company	
•	VENDOR to indicate Country of Origin.	

ATTACHMENT F - Inspection and Test Requirements

RFQ No:	PO No:	VENDOR:
Description and TAG No. MATERIAL REQUISITION FOR BARRED TEE		

NOTE: FINAL CERTIFIED DRAWINGS AND DOCUMENTS MUST BE SUBMITTED WITHIN TWO WEEKS OF FABRICATION AND TESTING COMPLETION

Table 7 - Inspection and Test Requirements

SCOPE: Barred Tee				Test and Inspection Requirements P. O. Specs and Respective Codes / Standards	
1 Visual Inspection Witnessing by EPC Contractor and Company (Note-1)					
2 Certificates / Records to be Checked by EPC Contractor & Owner (Note-1)					
3 Certificates / Data to Be Provided by VENDOR / Sub-Contractor					
01		X	X	01	Manufacturing procedure specification (MPS), Qualification Tests, as applicable
02		X	X	02	Material Test Reports, Certificate of Compliance for the Equipment
03	X	X	X	03	Positive Material testing, as applicable
04		X	X	04	Heat Treatment (Certificates), as Applicable
05		X	X	05	Heat Treatment (Charts), as Applicable
06		X	X	06	Welders Qualification Records, as Applicable
07		X	X	07	Welding Procedure Specifications, Procedure Qualification Records and Weldability Test, as Applicable
08	X	X	X	08	Workmanship, Material, Surface Defects, Components, Dimensions
09		X	X	09	Compliance with Purchase Order Description, Specifications, Certified Drawings, VENDORS Catalogue Description
10		X	X	10	Quality Control Manual - Check for System Implementation
11		X	X	11	Painting /Coating Inspection, as Applicable
					Non-Destructive Examinations (NDE)
12		X	X	12	Radiographic Testing of Welds, as Applicable
13		X	X	13	Magnetic Particle Examination of Bevel Ends, as Applicable
14		X	X	14	Ultrasonic Examination, as Applicable
15		X	X	15	Others as per Specifications

SCOPE: Barred Tee					Test and Inspection Requirements P. O. Specs and Respective Codes / Standards
1 Visual Inspection Witnessing by EPC Contractor and Company (Note-1)					
2 Certificates / Records to be Checked by EPC Contractor & Owner (Note-1)					
3 Certificates / Data to Be Provided by VENDOR / Sub-Contractor					
					Tests (As approved procedure)
16	X	X	X	16	Dimensional & Visual checks on all Barred Tee
17		X	X	17	100 % Calibration / Configuration checking
18	X	X	X	18	100 % Hydrostatic Testing
19		X	X	19	Testing of all other materials as per specification and all associated reference documents in this MR for more details
20	X	X	X	20	All key items and inspections and Tests as per the Project Specification & related Codes and Standards: 1. Testing of all Barred Tee 2. Mechanical Testing 3. Functional test operation 4. Painting / Protective Coating Inspection, etc.
					Other Requirements
21	X	X	X	21	Marking, Packing
22		X	X	22	Weighing
23		X	X	23	Compliance with Project Specification Requirements
24	X	X	X	24	Packing and Protection
25		X	X	25	Final Inspection and Issue of Release Note and Inspection Certification
26		X	X	26	Operating Manual, MRB, Installation and Maintenance manual
Inspection Level Required					I: II: X III: IV:
Notes: - VENDOR shall notify Company with min. (2) weeks in advance to witness the testing. - VENDOR to check the quality control requirement and Inspection Level in this table based on his experience, and further to advise, if there are deviations from VENDOR, VENDOR must seek Owner approval and provide the relevant reason. Legend: X- Required					

ATTACHMENT 1 - MTO for Barred Tee

Attachments	Document Title
ATTACHMENT 1.	MTO for Barred Tee



ATTACHMENT
1-MTO for Barred Te